Unit I : Analysis of Algorithms

Performance Analysis of an Algorithm – contd. :

Space Complexity : The space complexity of an algorithm is the amount of memory it needs to run to completion.

S(P) = c + Sp (instance characteristics) where,

c = space for (instructions, simple variables, fixed size aggregates)

Sp = variable part depends upon problem instance characteristics

Time Complexity : The time complexity of an algorithm is the amount of computer time it needs to run to completion.

T(P) = compile time + run time (execution time)

Compile time is constant or same program is executed many times, hence run time of a program (t_p) is important which depends upon instance characteristics

Performance Evaluation consists of :

1. A Priory Estimates or Performance or Asymptotic Analysis
2. A Posteriori Testing or Performance Measurement

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Performance Evaluation : Asymptotic analysis

Priori Estimates: For any instruction in a program:

Total Computation Time = Time required to execute the instruction * Number of times the instruction is executed in a program (i.e. **frequency count**).

Time required to execute the instruction depends upon:

- Speed of processor (Clock frequency)
- Instruction set of the machine language
- Time required (processor cycles) to execute an instruction
- Compiler used to translate high level instructions to m/c language

All above criteria differ from installation to installation. Therefore for priori estimates frequency count of each statement is most important factor while analyzing an algorithm or program.

Frequency count : It is the number of times an instruction is executed in the execution of a program. Example